a Projecting scenario-consistent Local Climate Zones

Estimating demand · how much of each LCZ?

Processes 1–2

Global land-use data
1 km · 2020–2100

Urban-fraction data
1 km · 2020–2100

1 · Integration by SSP

Major land-use fractions
2020–2100

Historical LCZ maps
100 m · 1986–2016

2 · Markov chain analysis

Projected LCZ composition within each major land-use category

LCZ transition matrix

LCZ demand projections

Allocating change in space · PLUS · Processes 3–5

3 · Land expansion analysis (LEAS)

LCZ expansion maps

4 · Random forest model

Growth probabilities

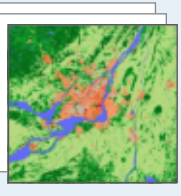
5 · Cellular automaton with random patch seeds

Driving factors
25 variables · GeoIndex

Influence of each driver on LCZ expansion

Validation: 2001 → 2016 hindcast
FoM 0.33–0.36 · OA 0.81–0.83

Decadal LCZ maps
5 SSPs · 3 cities
2030–2100



LCZ classes (WUDAPT)

1 2 3 4 5 6 7 8 9 10

A B C D E F G

b Downscaling to 1-km climate

Projected LCZ surfaces
dynamic SSPs + static control

Surface parameters
WUDAPT → WRF (w2w)

CASPER
U-Net WRF emulator
bias-corrected CESM forcing

1-km fields · 6-hourly
T2 · skin temperature · humidity · wind

264 runs = 3 cities × (8 SSP–RCP pairs + static under 3 RCPs) × 8 decades
Climate diagnostics: UHI, humidex and ventilation · same forcing in each static–dynamic pair

Blue border: input data · Square: numbered process · Rounded: product · Filled: core model